



Graph Neural Networks Applications

Emanuel Sallinger



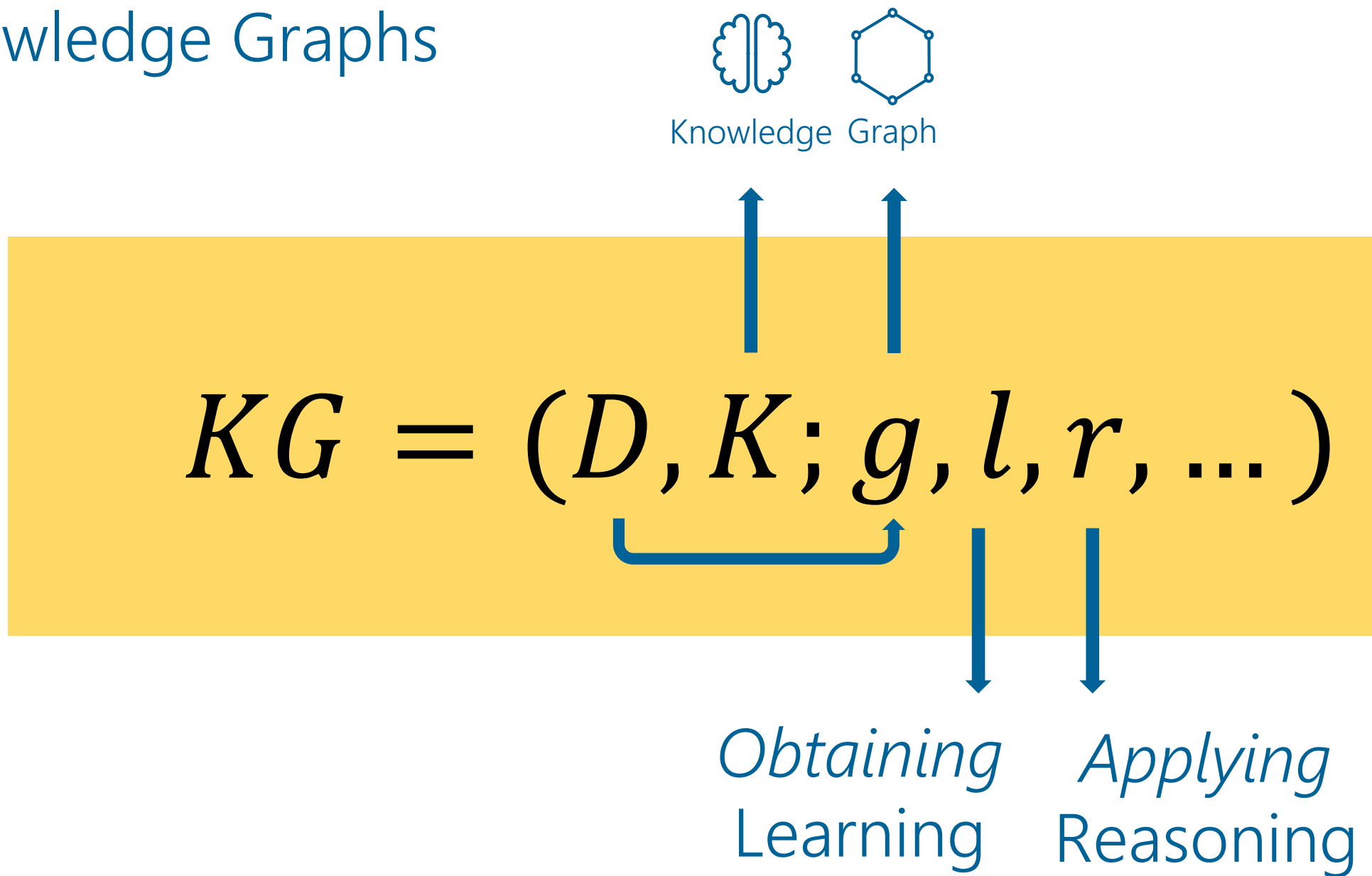
Graph Neural Networks

$$\mathbf{H}^{t+1} = F(\mathbf{H}^t, \mathbf{X})$$

F propagation step



Knowledge Graphs





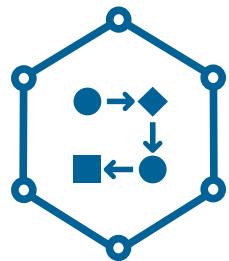
Graph Neural Networks: **Latent Knowledge**

$$KG = (D, K; g, l, r)$$

$$\mathbb{K} = \mathbb{W} \text{ (w.r.t. } F)$$

$$l: \mathbb{D} \rightarrow \mathbb{W}$$

$$r_{\mathbb{W}}: \mathbb{D} \rightarrow \mathbb{D}$$



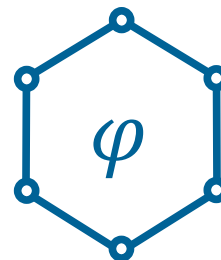
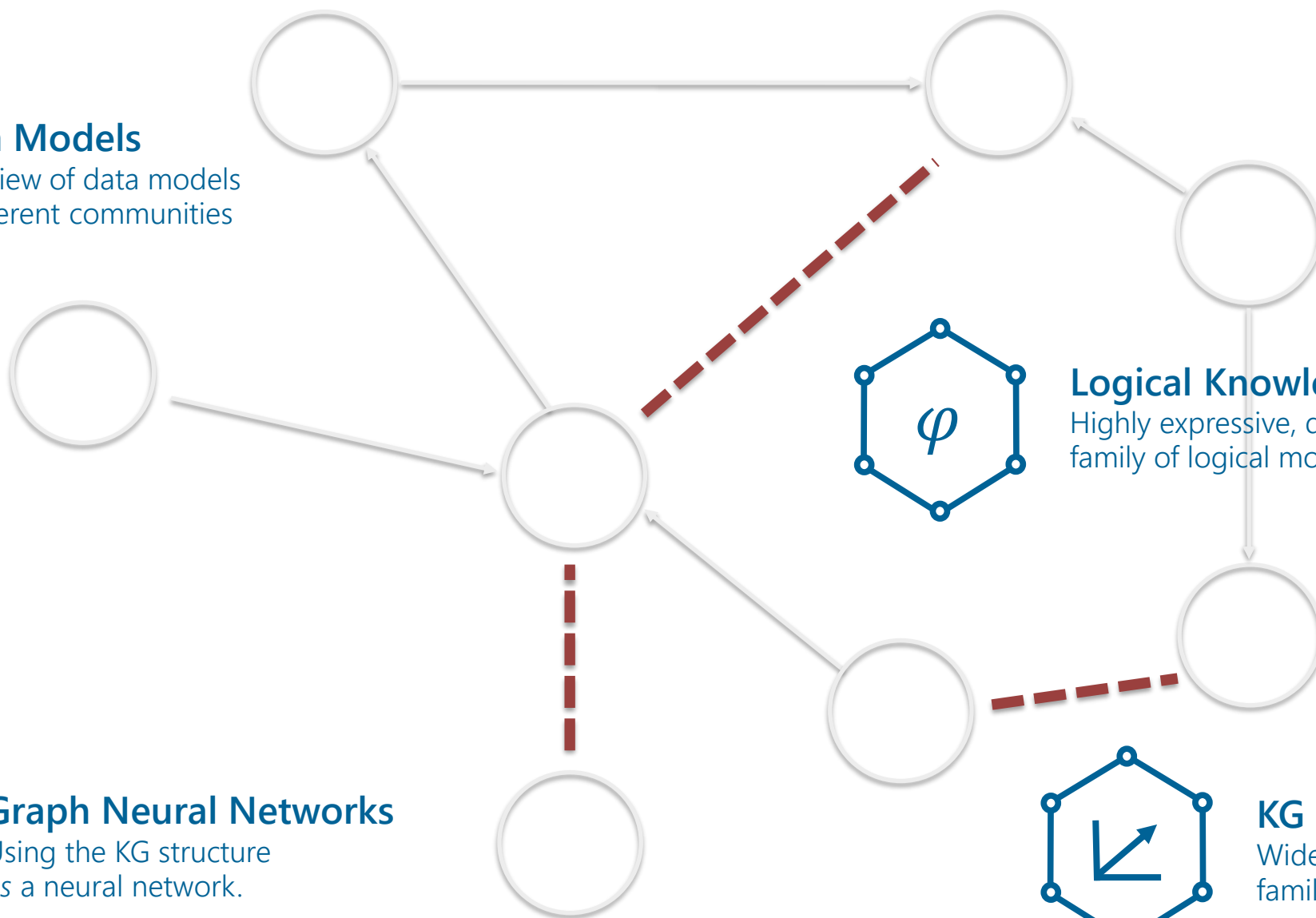
Data Models

Overview of data models in different communities



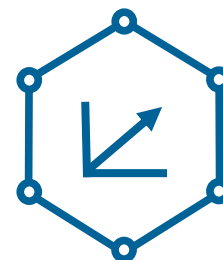
Graph Neural Networks

Using the KG structure as a neural network.



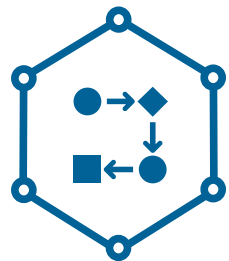
Logical Knowledge in KGs

Highly expressive, diverse family of logical models.



KG Embeddings

Widely-applied, large family of ML models



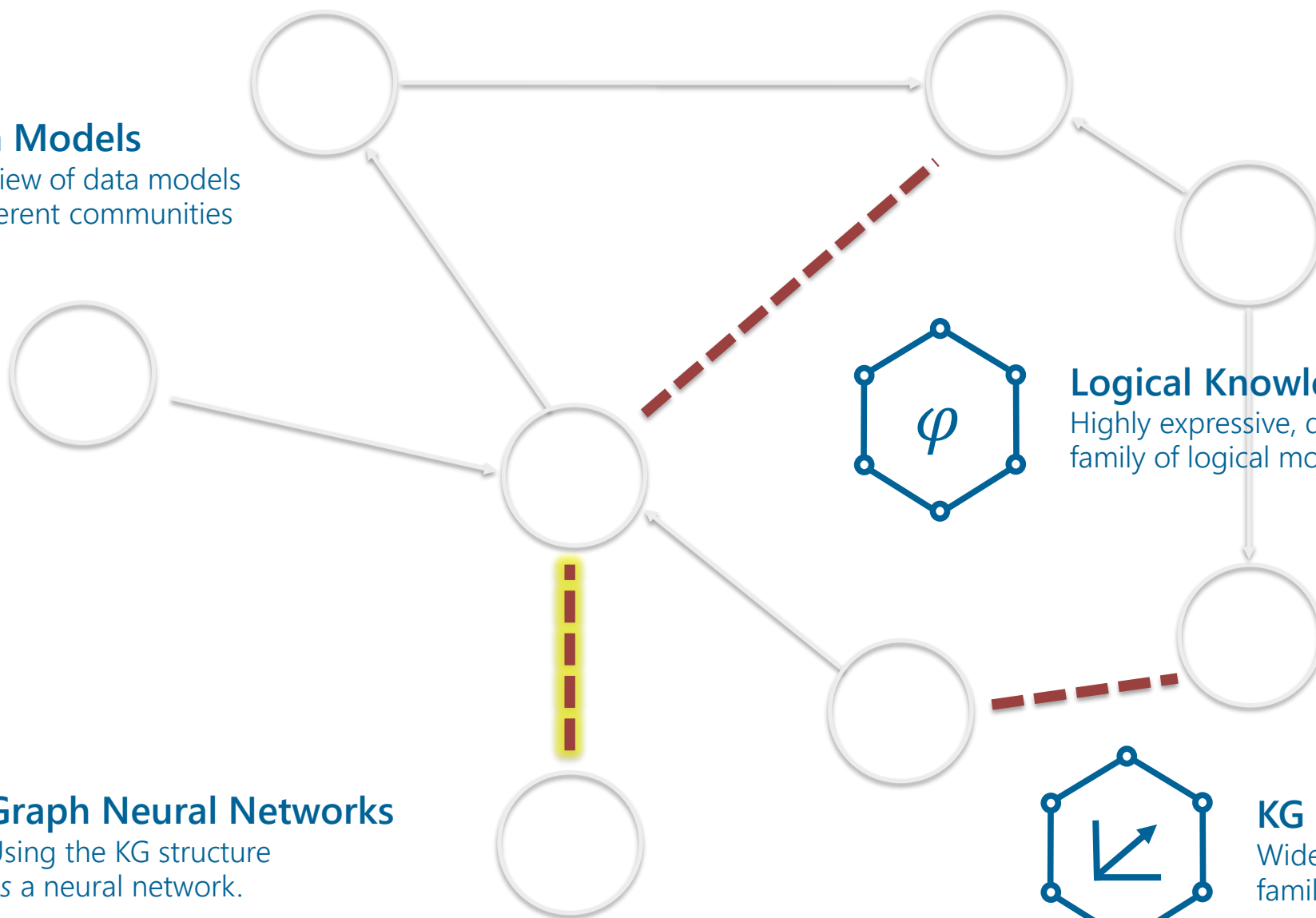
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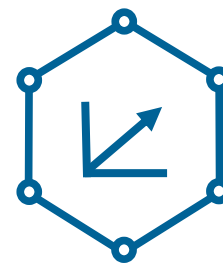
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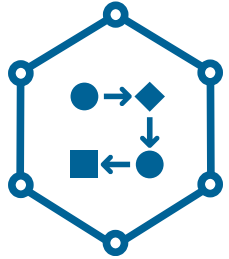
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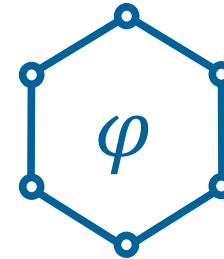
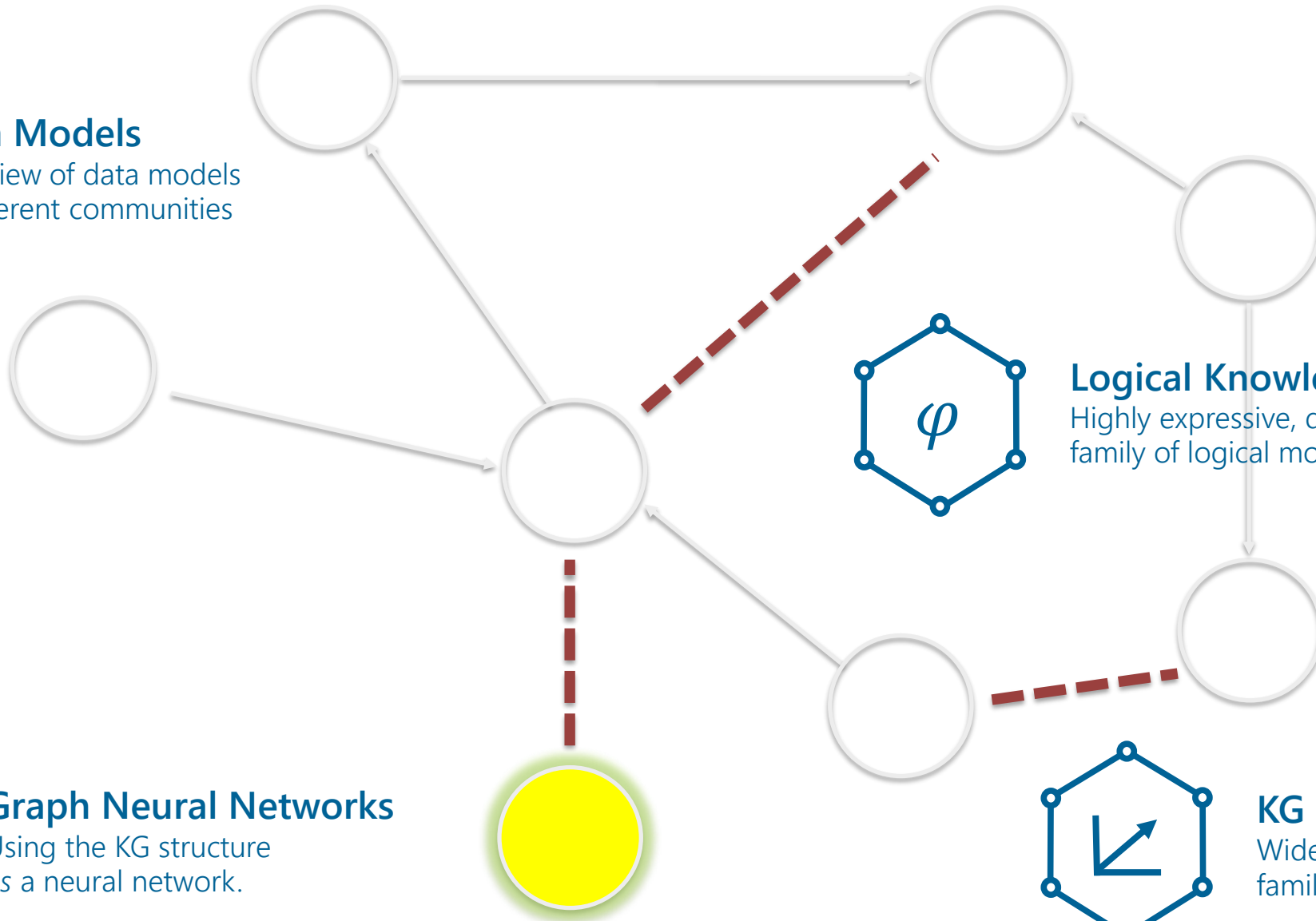
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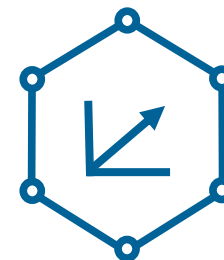
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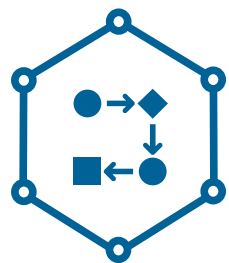
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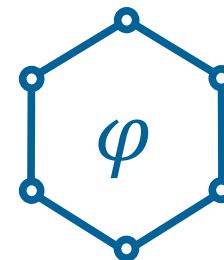
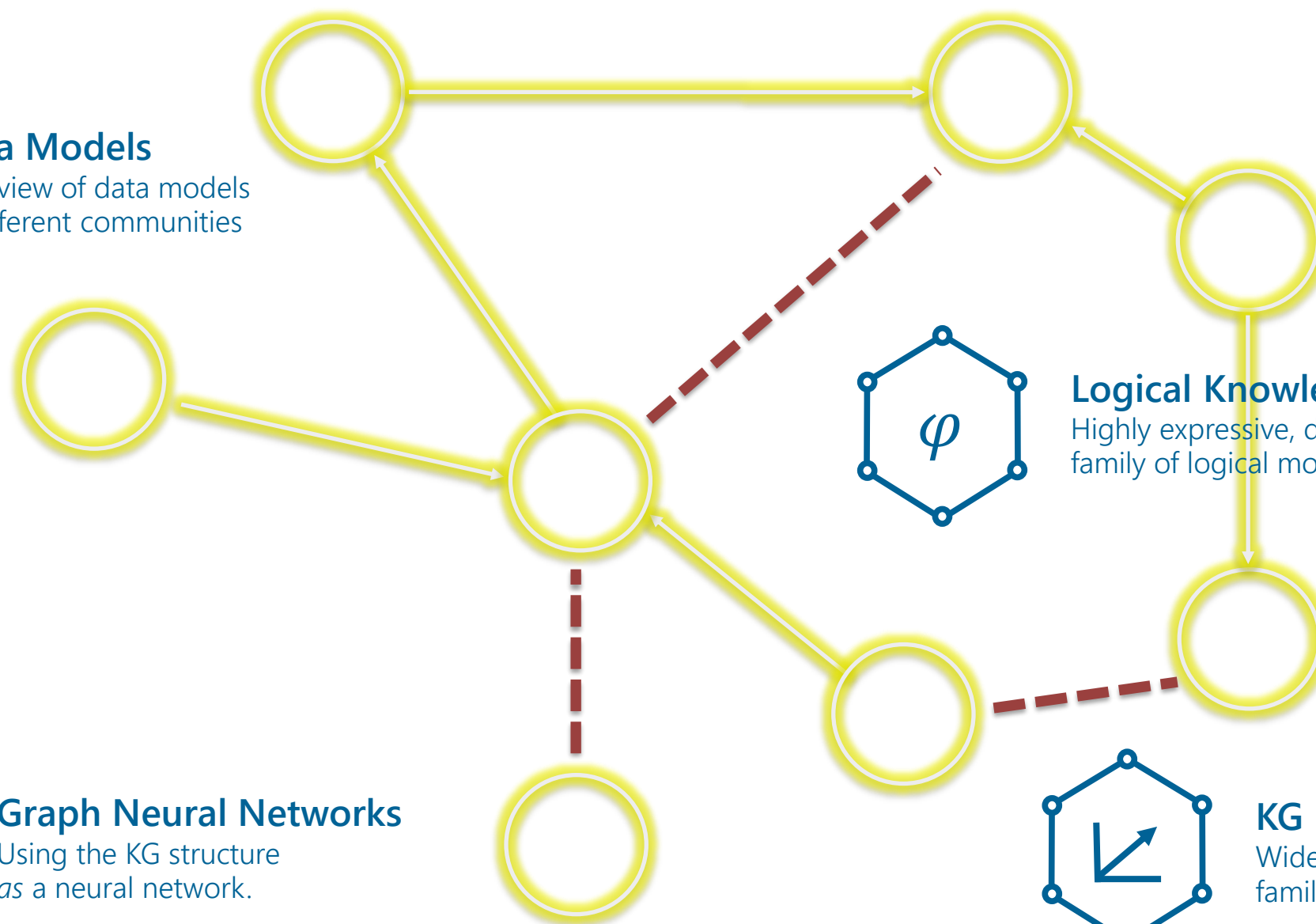
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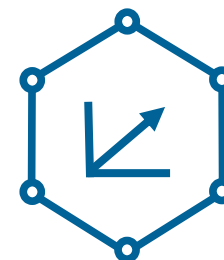
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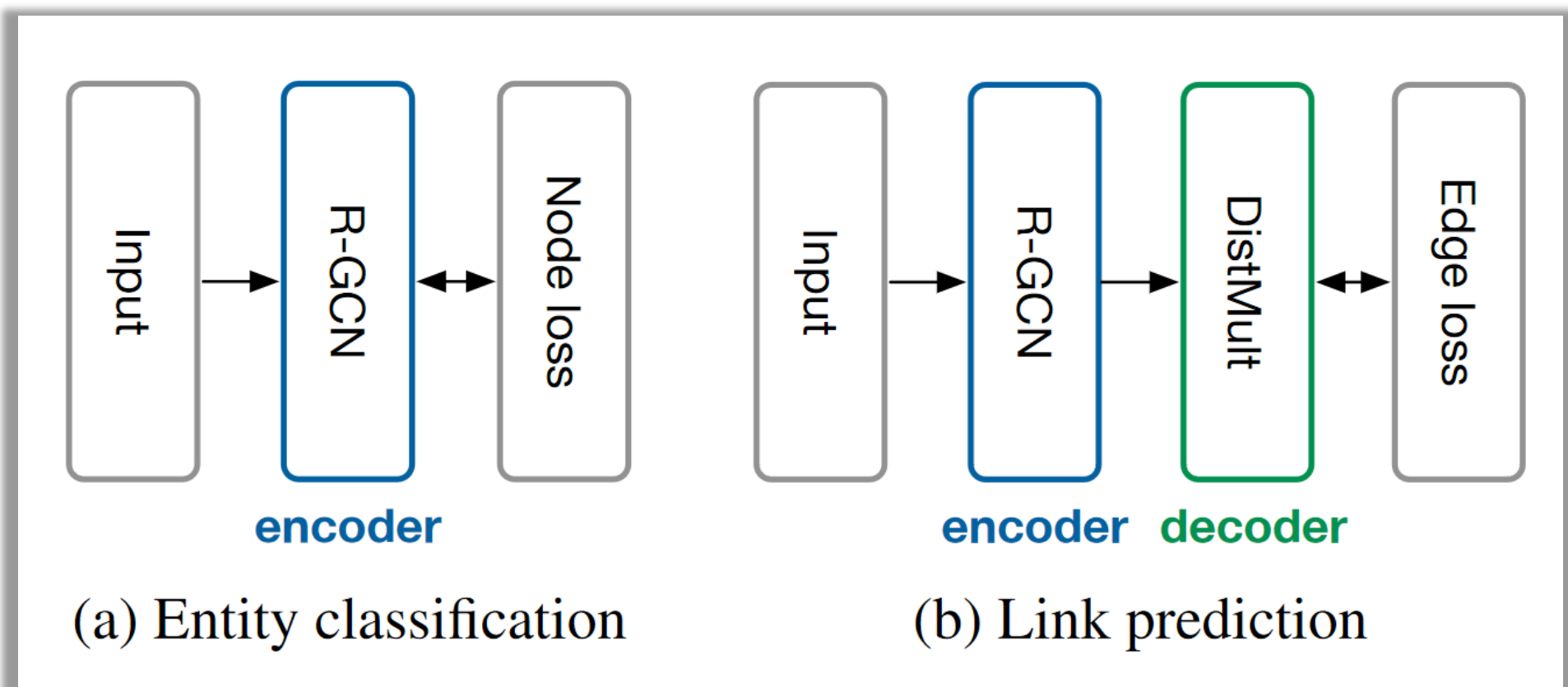


KG Embeddings

Widely-applied, large family of ML models

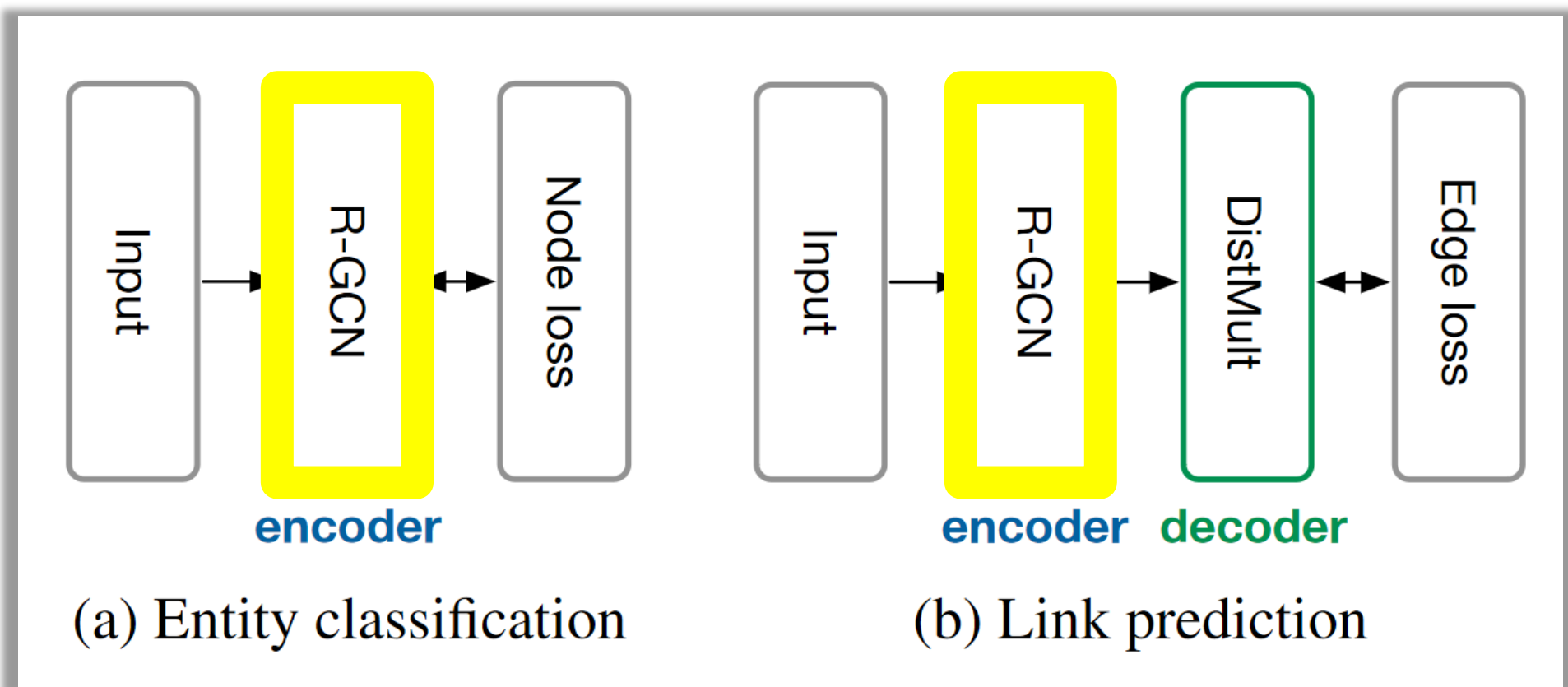


Link Prediction





Link Prediction





Link Prediction

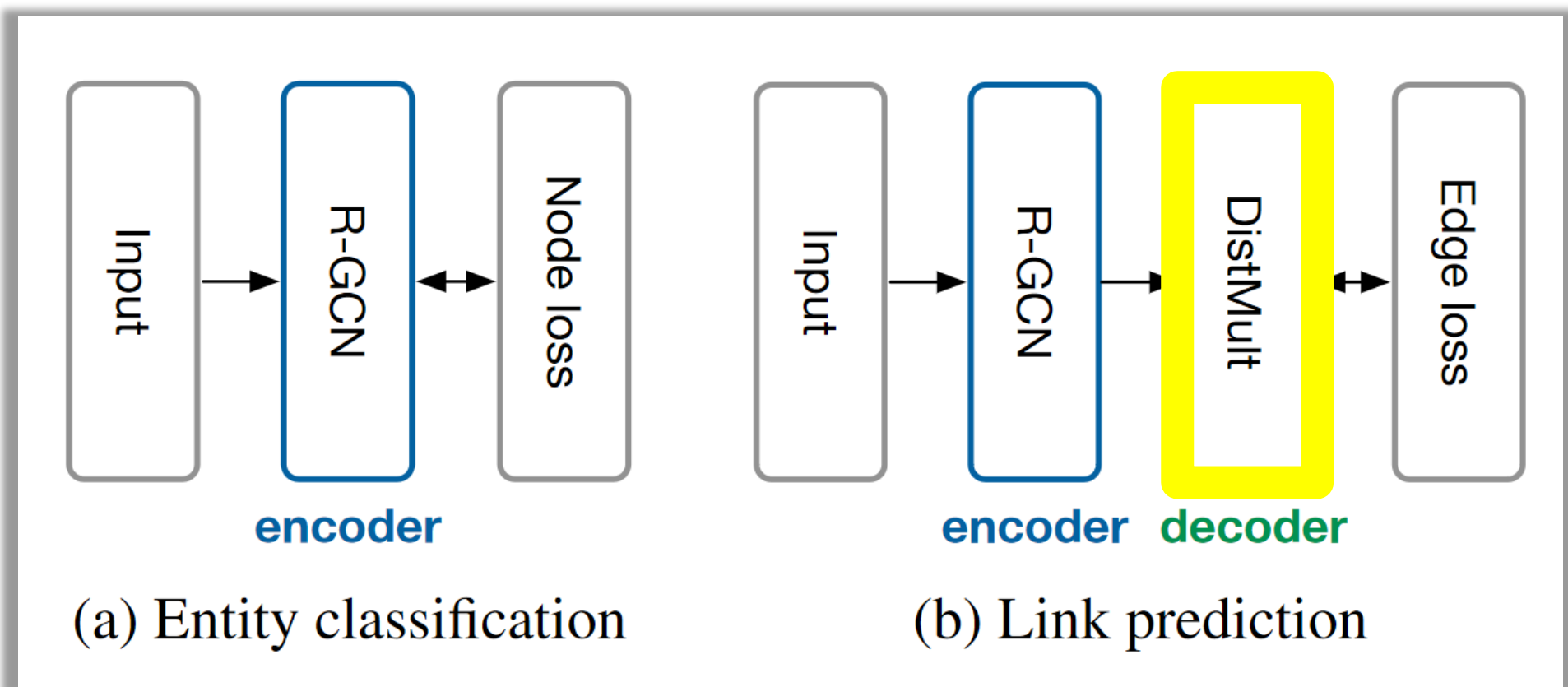




Table 3
Applications of graph neural networks.

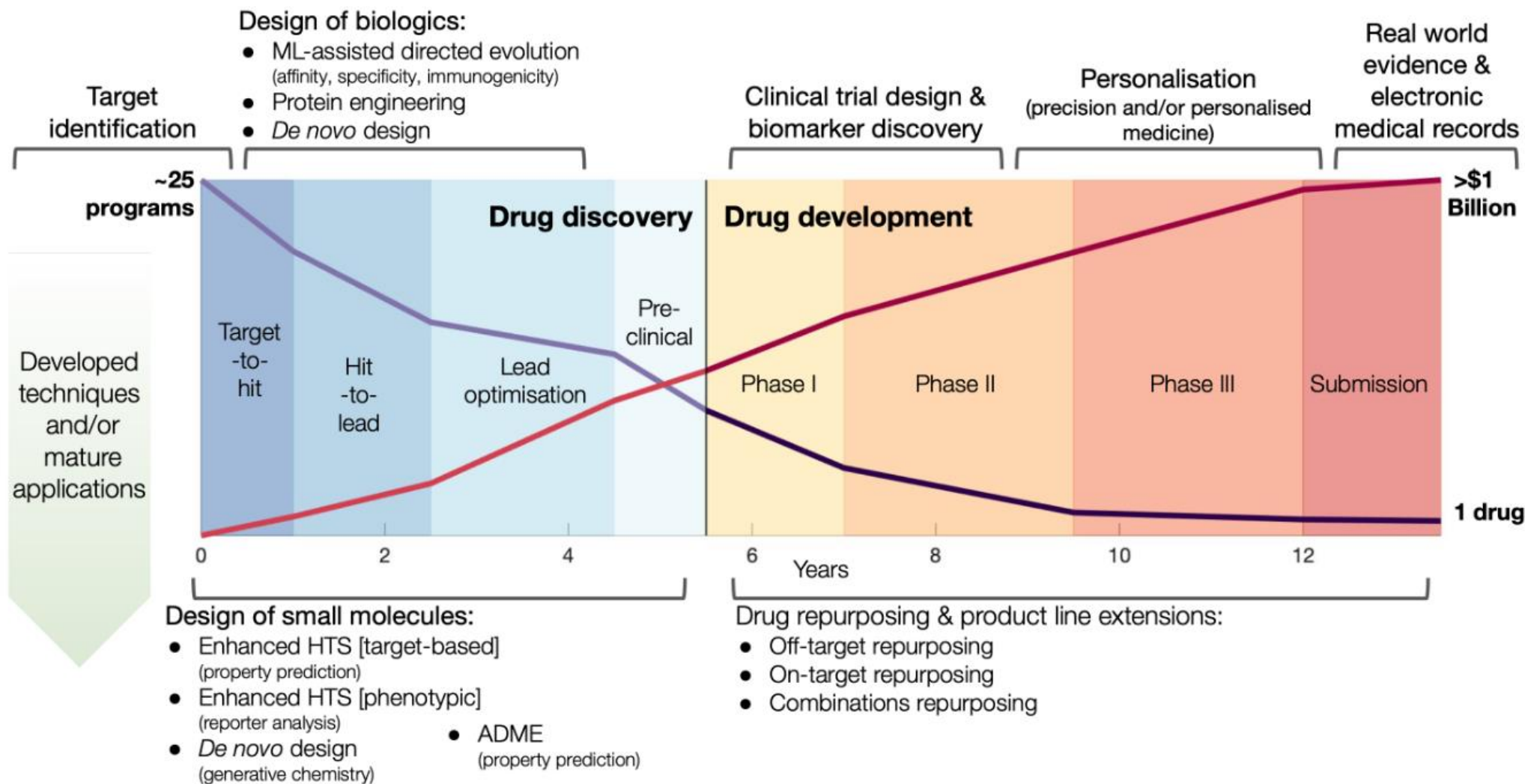
Area	Application	References
Graph Mining	Graph Matching	(Riba et al., 2018; Li et al., 2019b)
	Graph Clustering	(Zhang et al., 2019c; Ying et al., 2018b; Tsitsulin et al., 2020)
Physics	Physical Systems Modeling	(Battaglia et al., 2016; Sukhbaatar Fergus et al., 2016; Watters et al., 2017; Hoshen, 2017; Kipf et al., 2018; Sanchez et al., 2018)
Chemistry	Molecular Fingerprints Chemical Reaction Prediction	(Duvenaud et al., 2015; Kearnes et al., 2016) Do et al. (2019)
Biology	Protein Interface Prediction Side Effects Prediction Disease Classification	Fout et al. (2017) Zitnik et al. (2018) Rhee et al. (2018)
Knowledge Graph	KB Completion KG Alignment	(Hamaguchi et al., 2017; Schlichtkrull et al., 2018; Shang et al., 2019) (Wang et al., 2018b; Zhang et al., 2019d; Xu et al., 2019c)
Generation	Graph Generation	(Shchur et al., 2018b; Nowak et al., 2018; Ma et al., 2018; You et al., 2018a, 2018b; De Cao and Kipf, 2018; Li et al., 2018d; Shi et al., 2020; Liu et al., 2019; Grover et al., 2019)
Combinatorial Optimization	Combinatorial Optimization	(Khalil et al., 2017; Nowak et al., 2018; Li et al., 2018; Kool et al., 2019; Bello et al., 2017; Vinyals et al., 2015b; Sutton and Barto, 2018; Dai et al., 2016; Gasse et al., 2019; Zheng et al., 2020a; Selsam et al., 2019; Sato et al., 2019)
Traffic Network	Traffic State Prediction	(Cui et al., 2018b; Yu et al., 2018; Zheng et al., 2020b; Guo et al., 2019)
Recommendation Systems	User-item Interaction Prediction	(van den Berg et al., 2017; Ying et al., 2018a)
	Social Recommendation	(Wu et al., 2019c; Fan et al., 2019)
Others (Structural)	Stock Market Software Defined Networks AMR Graph to Text	(Matsunaga et al., 2019; Yang et al., 2019; Chen et al., 2018c; Li et al., 2020; Kim et al., 2019) Rusek et al. (2019) (Song et al., 2018a; Beck et al., 2018)
Text	Text Classification Sequence Labeling Neural Machine Translation Relation Extraction Event Extraction Fact Verification Question Answering Relational Reasoning	(Peng et al., 2018; Yao et al., 2019; Zhang et al., 2018d; Tai et al., 2015) (Zhang et al., 2018d; Marcheggiani and Titov, 2017) (Bastings et al., 2017; Marcheggiani et al., 2018; Beck et al., 2018) (Miwa and Bansal, 2016; Peng et al., 2017; Song et al., 2018b; Zhang et al., 2018f) (Nguyen and Grishman, 2018; Liu et al., 2018) (Zhou et al., 2019; Liu et al., 2020; Zhong et al., 2020) (Song et al., 2018c; De Cao et al., 2019; Qiu et al., 2019; Tu et al., 2019; Ding et al., 2019) (Santoro et al., 2017; Palm et al., 2018; Battaglia et al., 2016)
Image	Social Relationship Understanding	Wang et al. (2018c)
	Image Classification	(Garcia and Bruna, 2018; Wang et al., 2018d; Lee et al., 2018b; Kampffmeyer et al., 2019; Marino et al., 2017)
	Visual Question Answering	(Teney et al., 2017; Wang et al., 2018c; Narasimhan et al., 2018)
	Object Detection	(Hu et al., 2018; Gu et al., 2018)
	Interaction Detection	(Qi et al., 2018; Jain et al., 2016)
	Region Classification	Chen et al. (2018d)
Other (Non-structural)	Semantic Segmentation	(Liang et al., 2016, 2017; Landrieu and Simonovsky, 2018; Wang et al., 2018e; Qi et al., 2017b)
	Program Verification	(Allamanis et al., 2018; Li et al., 2016)

Source: <https://arxiv.org/ftp/arxiv/papers/1812/1812.08434.pdf>

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	Disease Classification	Rhee et al. (2018)





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	Software Defined Networks	Rusek et al. (2019)
	AMR Graph to Text	(Song et al., 2018a; Beck et al., 2018)
Text	Text Classification	(Peng et al., 2018; Yao et al., 2019; Zhang et al., 2018d; Tai et al., 2015)
	Sequence Labeling	(Zhang et al., 2018d; Marcheggiani and Titov, 2017)
	Neural Machine Translation	(Bastings et al., 2017; Marcheggiani et al., 2018; Beck et al., 2018)
	Relation Extraction	(Miwa and Bansal, 2016; Peng et al., 2017; Song et al., 2018b; Zhang et al., 2018f)
	Event Extraction	(Nguyen and Grishman, 2018; Liu et al., 2018)



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